

Post_hoc

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ANOVA clean workflow

```
# Load Data & Prepare
data <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/One-Way-ANOVA/1.csv")
data$Treatment <- as.factor(data$Treatment) # ensure categorical variable

#Build ANOVA Model
model <- aov(Yield ~ Treatment, data = data)

#Check Assumptions
##Normality of Residuals
shapiro.test(residuals(model))
```

Shapiro-Wilk normality test

```
data: residuals(model)
W = 0.93739, p-value = 0.3183
```

Interpretation

$p > 0.05 \rightarrow$ acceptable (not perfect, just no strong violation)

$p < 0.05 \rightarrow$ check plots before decision

```
## Homogeneity of Variance
```

```
library(car)
```

Warning: package 'car' was built under R version 4.4.3

Loading required package: carData

```
leveneTest(Yield ~ Treatment, data = data)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group 3    0.75 0.5431
      12
```

Interpretation

$p > 0.05 \rightarrow$ equal variance (good)

$p < 0.05 \rightarrow$ violation (be careful)

Decision Rule (Important)

If assumptions roughly OK $\rightarrow$ proceed with ANOVA

If strong violation $\rightarrow$ go to non-parametric (Kruskal)

```
#Run ANOVA
summary(model)
```

```
      Df Sum Sq Mean Sq F value    Pr(>F)
Treatment    3  642.8   214.25   233.7 6.61e-11 ***
Residuals   12    11.0     0.92
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

$p < 0.05 \rightarrow$ reject null $\rightarrow$ at least one group differs $\rightarrow$ go for post hoc

$p > 0.05 \rightarrow$ fail to reject $\rightarrow$ stop (no post hoc)

Post Hoc Tests

1. Tukey's HSD (BEST CHOICE)

```
library(agricolae)
```

Warning: package 'agricolae' was built under R version 4.4.3

```
tukey_result <- HSD.test(model, "Treatment", group = TRUE) # if group is TRUE, it will show t
tukey_result
```

\$statistics

MSerror	Df	Mean	CV	MSD
0.9166667	12	21.625	4.427409	2.009956

\$parameters

test	name.t	ntr	StudentizedRange	alpha
Tukey	Treatment	4	4.19866	0.05

\$means

	Yield	std	r	se	Min	Max	Q25	Q50	Q75
T1	13.5	1.2909944	4	0.4787136	12	15	12.75	13.5	14.25
T2	18.0	0.8164966	4	0.4787136	17	19	17.75	18.0	18.25
T3	25.0	0.8164966	4	0.4787136	24	26	24.75	25.0	25.25
T4	30.0	0.8164966	4	0.4787136	29	31	29.75	30.0	30.25

\$comparison

NULL

\$groups

	Yield	groups
T4	30.0	a
T3	25.0	b
T2	18.0	c

```
T1 13.5      d
```

```
attr(,"class")  
[1] "group"
```

Interpretation

- \$means -> show the mean yield per fertilizer
- \$comparison -> signif (stars indicate significance) ..*** = highly significant ($p < 0.001$)
if ** = $p < 0.01$ if * = $p < 0.05$
- \$Group -> Same letter= no significant difference, Different letters = significantly different

Show the specific result:

```
tukey_result$means  
tukey_result$groups  
tukey_result$comparison
```

Why best:

controls error properly accepted in research papers

Use when:

- 3+ groups final report / publication
- Used mainly in agriculture papers

2. LSD Test (Least Strict)

```
library(agricolae)  
lsd_result <- LSD.test(model, "Treatment", p.adj = "none")  
lsd_result
```

```
$statistics  
  MSerror Df   Mean      CV t.value    LSD  
0.9166667 12 21.625 4.427409 2.178813 1.475063
```

```
$parameters  
      test p.adjusted  name.t ntr alpha  
Fisher-LSD      none Treatment  4  0.05
```

```
$means
      Yield      std r      se      LCL      UCL Min Max   Q25   Q50   Q75
T1   13.5  1.2909944  4 0.4787136 12.45697 14.54303  12  15 12.75 13.5 14.25
T2   18.0  0.8164966  4 0.4787136 16.95697 19.04303  17  19 17.75 18.0 18.25
T3   25.0  0.8164966  4 0.4787136 23.95697 26.04303  24  26 24.75 25.0 25.25
T4   30.0  0.8164966  4 0.4787136 28.95697 31.04303  29  31 29.75 30.0 30.25
```

```
$comparison
NULL
```

```
$groups
      Yield groups
T4   30.0      a
T3   25.0      b
T2   18.0      c
T1   13.5      d
```

```
attr(,"class")
[1] "group"
```

Same interpretation as Tukey.

LSD limitation:

LSD test is more liberal (less strict), so it's: Good for few treatments (2–4), exploratory / early analysis. You want high sensitivity (detect small differences)

Rule of Thumb

Not good for many groups → higher chance of false positives.

- 2–3 groups → LSD is OK, especially for early or internal analysis
- 3+ groups or final result/report → Prefer Tukey's HSD
- If unsure, go with Tukey — it's safer and accepted

Use LSD when

- 2–3 treatments only
- exploratory / early analysis
- you want high sensitivity (detect small differences)

Problem

high false positives. That is why go with Tukey.

3. Duncan's Test (Middle Ground)

```
library(agricolae)
duncan_result <- duncan.test(model, "Treatment", group = TRUE)
duncan_result
```

\$statistics

MSerror	Df	Mean	CV
0.9166667	12	21.625	4.427409

\$parameters

test	name.t	ntr	alpha
Duncan	Treatment	4	0.05

\$duncan

	Table	CriticalRange
2	3.081307	1.475063
3	3.225244	1.543968
4	3.312453	1.585716

\$means

	Yield	std	r	se	Min	Max	Q25	Q50	Q75
T1	13.5	1.2909944	4	0.4787136	12	15	12.75	13.5	14.25
T2	18.0	0.8164966	4	0.4787136	17	19	17.75	18.0	18.25
T3	25.0	0.8164966	4	0.4787136	24	26	24.75	25.0	25.25
T4	30.0	0.8164966	4	0.4787136	29	31	29.75	30.0	30.25

\$comparison

NULL

\$groups

	Yield	groups
T4	30.0	a
T3	25.0	b
T2	18.0	c
T1	13.5	d

```
attr("class")
[1] "group"
```

Characteristics

- more sensitive than Tukey
- less strict → more risky

Alternative using emmeans library(emmeans)

LSD equivalent

```
library(emmeans)
```

Warning: package 'emmeans' was built under R version 4.4.3

Welcome to emmeans.

Caution: You lose important information if you filter this package's results.

See '? untidy'

```
emmeans(model, pairwise ~ Treatment, adjust = "none")
```

\$emmeans

Treatment	emmean	SE	df	lower.CL	upper.CL
T1	13.5	0.479	12	12.5	14.5
T2	18.0	0.479	12	17.0	19.0
T3	25.0	0.479	12	24.0	26.0
T4	30.0	0.479	12	29.0	31.0

Confidence level used: 0.95

\$contrasts

contrast	estimate	SE	df	t.ratio	p.value
T1 - T2	-4.5	0.677	12	-6.647	<.0001
T1 - T3	-11.5	0.677	12	-16.987	<.0001
T1 - T4	-16.5	0.677	12	-24.372	<.0001
T2 - T3	-7.0	0.677	12	-10.340	<.0001
T2 - T4	-12.0	0.677	12	-17.725	<.0001
T3 - T4	-5.0	0.677	12	-7.385	<.0001

Tukey

```
library(emmeans)
emmeans(model, pairwise ~ Treatment, adjust = "tukey")
```

```
$emmeans
  Treatment emmean      SE df lower.CL upper.CL
T1          13.5 0.479 12     12.5     14.5
T2          18.0 0.479 12     17.0     19.0
T3          25.0 0.479 12     24.0     26.0
T4          30.0 0.479 12     29.0     31.0
```

Confidence level used: 0.95

```
$contrasts
 contrast estimate      SE df t.ratio p.value
T1 - T2      -4.5 0.677 12  -6.647 0.0001
T1 - T3     -11.5 0.677 12 -16.987 <.0001
T1 - T4     -16.5 0.677 12 -24.372 <.0001
T2 - T3      -7.0 0.677 12 -10.340 <.0001
T2 - T4     -12.0 0.677 12 -17.725 <.0001
T3 - T4      -5.0 0.677 12  -7.385 <.0001
```

P value adjustment: tukey method for comparing a family of 4 estimates

Difference Between the Test:

1. **Sensitivity** = How easily the test finds a difference

- High sensitivity = Finds small differences
- Low sensitivity = Only finds big differences

2. **Strictness** = How careful the test is to avoid wrong results

- Strict test = Needs more evidence to say “groups are different”
- Less strict = Can say “different” even with less evidence (risky)

Final Summary Table:

Test	When use	Sensitivity	Strictness	Error Control	Most Used
Turkey	Safe, journal standard, >3 groups	Lower	High	Strong	Yes
LSD	2-3 groups, quick idea, early research	High	Low	None	Less
Duncan	Grouping letters, medium strictness	High	Medium	Some	Some Agri

Dunn Test(non-parametric)

If Assumptions FAIL → Non-Parametric Path (Kruskal-Wallis Test)

```
kruskal.test(Yield ~ Treatment, data = data)
```

Interpretation

$p < 0.05$ → groups differ → go for **Dunn's test**

```
library(FSA)
```

Registered S3 methods overwritten by 'FSA':

```
method      from
confint.boot car
hist.boot    car
```

```
## FSA v0.10.0. See citation('FSA') if used in publication.
```

```
## Run fishR() for related website and fishR('IFAR') for related book.
```

Attaching package: 'FSA'

The following object is masked from 'package:car':

```
bootCase
```

```
dunnTest(Yield ~ Treatment, data = data, method = "bonferroni")
```

Dunn (1964) Kruskal-Wallis multiple comparison

p-values adjusted with the Bonferroni method.

	Comparison	Z	P.unadj	P.adj
1	T1 - T2	-1.190807	0.2337294644	1.0000000000
2	T1 - T3	-2.381613	0.0172369795	0.103421877
3	T2 - T3	-1.190807	0.2337294644	1.0000000000
4	T1 - T4	-3.572420	0.0003536973	0.002122184
5	T2 - T4	-2.381613	0.0172369795	0.103421877
6	T3 - T4	-1.190807	0.2337294644	1.0000000000

Interpretation

$p_{\text{adj}} < 0.05 \rightarrow$ significant difference.